MTH622:Quiz no.4

Quiz Start Time: 04:50 PM, 10 March 2022

Question # 2 of 5 (Start time: 04:50:38 PM, 10 March 2022)

Total Marks: 1

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate systems, then $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$ where it is not necessary that the origins of the two systems coincide initially.

Select the correct option

Reload Math Equations

- false
- true



Click to Save Answer & Move to Next Question

MTH622:Quiz no.4

Quiz Start Time: 04:50 PM, 10 March 2022

Question # 4 of 5 (Start time: 04:52:26 PM, 10 March 2022)

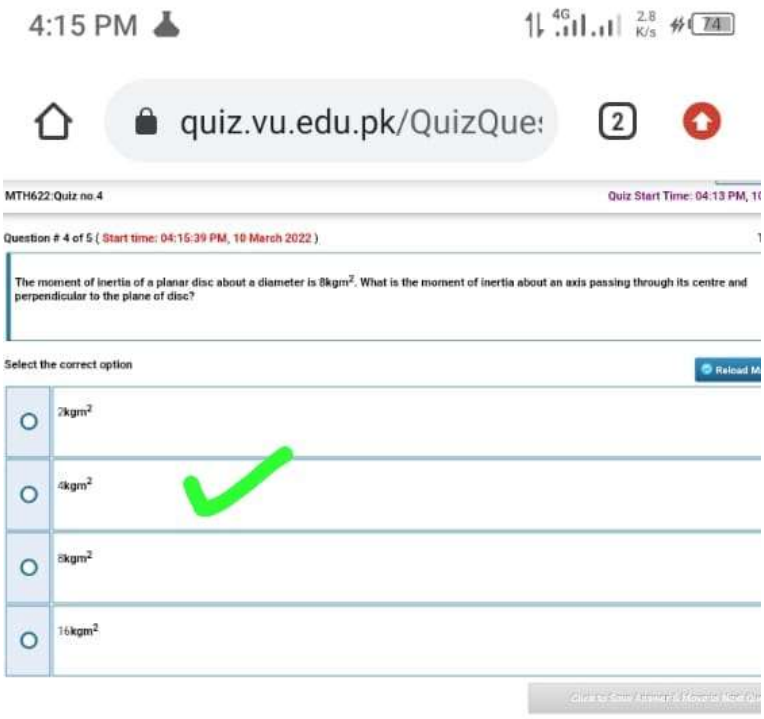
Total Marks: 1

The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus
the moment of inertia due to the total mass placed at the centroid

Select the correct option

- true
- false

Click to Save / Answer & Move to Next Question


4:15 PM
quiz.vu.edu.pk/QuizQue
MTH622:Quiz no.4
Quiz Start Time: 04:13 PM, 1
Question # 4 of 5 (Start time: 04:15:39 PM, 10 March 2022)
The moment of inertia of a planar disc about a diameter is 8kgm². What is the moment of inertia about an axis passing through its centre and perpendicular to the plane of disc?
Select the correct option
2kgm²
4kgm²
8kgm²
16kgm²

4:13 PM

quiz.vu.edu.pk/QuizQue:

MTH622:Quiz no.4

Quiz Start Time: 04:13 PM,

Question # 2 of 5 (Start time: 04:13:45 PM, 10 March 2022)

Three particles of masses m, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find I_{xx}

Select the correct option

- 9m
- 8m

Click to Save Answer & Move to Next Q

3:56 PM

quiz.vu.edu.pk

TH622:Quiz no.4 Quiz Start Time: 03:56 PM, 1

estion # 1 of 5 (Start time: 03:56:38 PM, 10 March 2022)

Three particles of masses m, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find I_{zz}

lect the correct option

Reload

8m

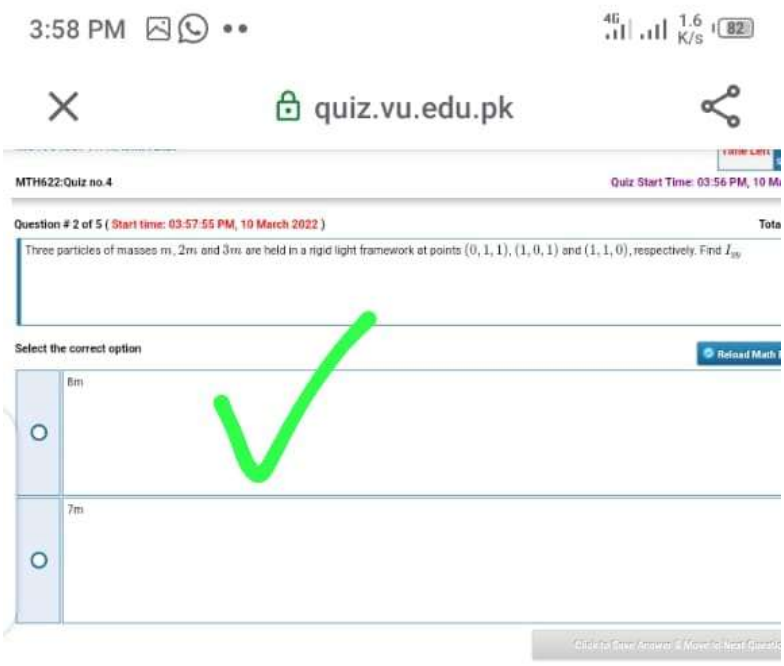
I_{yy}

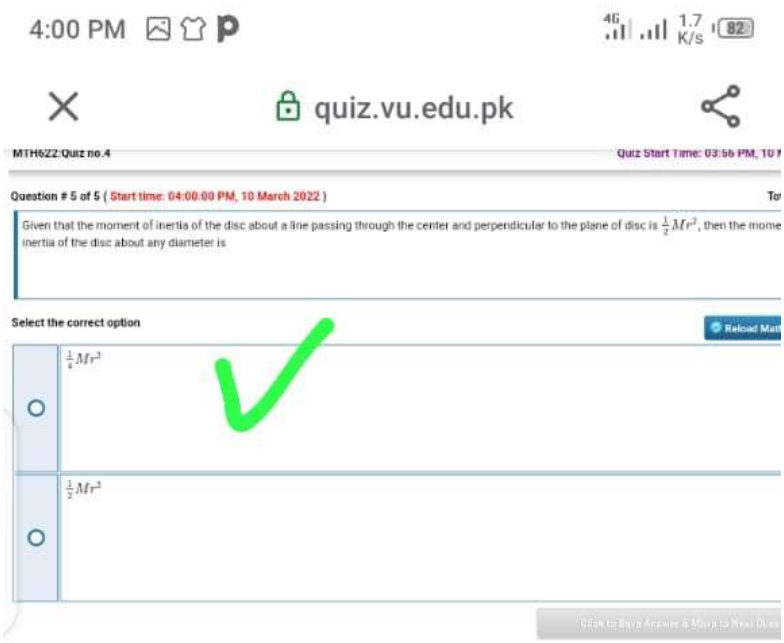
7m

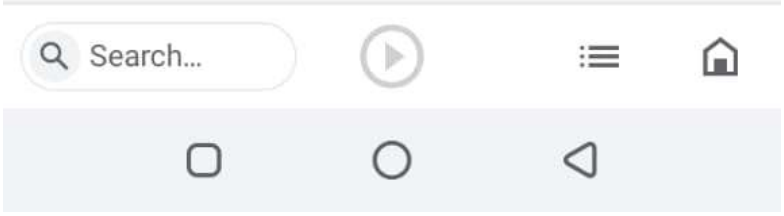
$\checkmark I_{xx}$

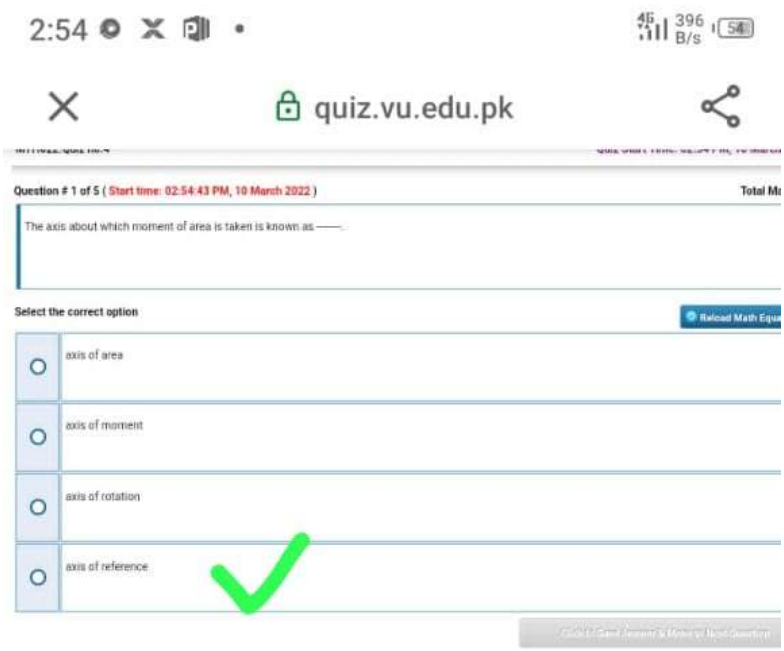
Click to Save Answer & Move to Next Qu

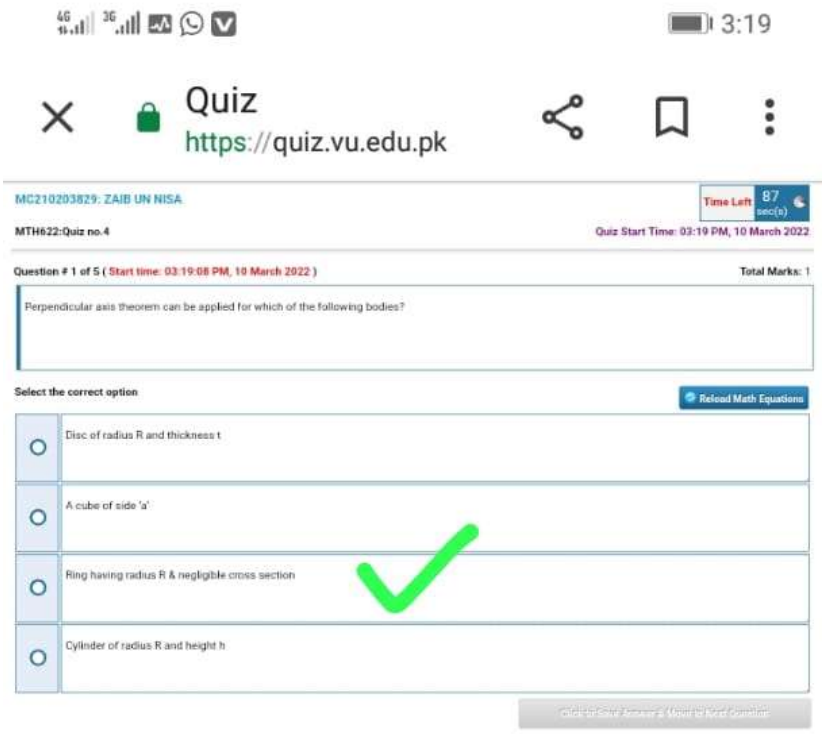
Search...

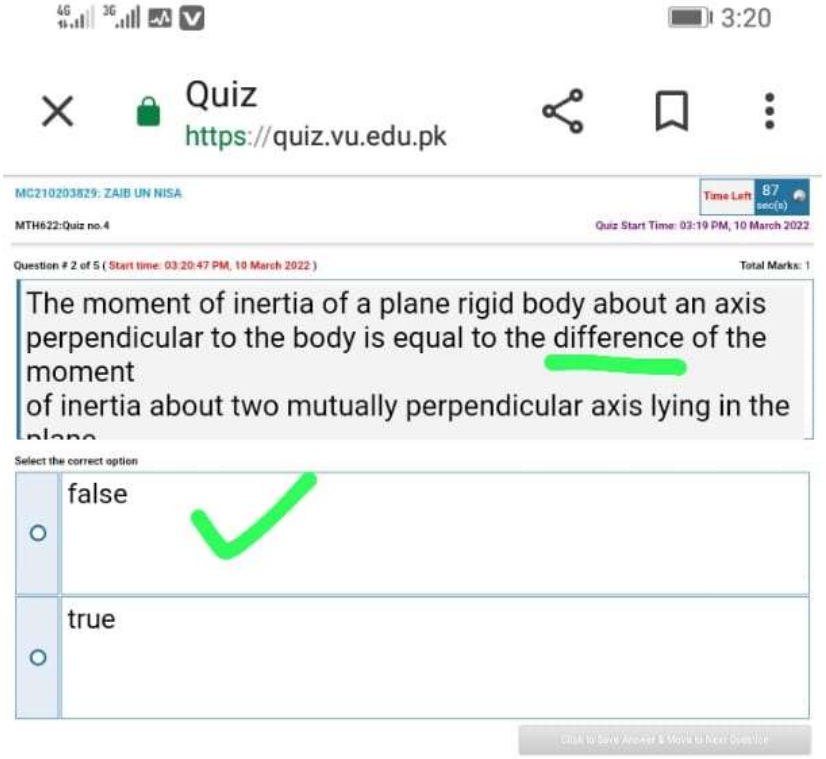

3:58 PM
1.6 K/s
82
quiz.vu.edu.pk
MTH622:Quiz no.4
Quiz Start Time: 03:56 PM, 10 M
Question # 2 of 5 (Start time: 03:57:55 PM, 10 March 2022)
Tota
Three particles of masses m, 2m and 3m are held in a rigid light framework at points (0, 1, 1), (1, 0, 1) and (1, 1, 0), respectively. Find I_{xy}
Select the correct option
Reload Math
8m
7m


4:00 PM
1.7 K/s
82
quiz.vu.edu.pk
MTH6ZZ:Quiz no.4
Quiz Start Time: 03:56 PM, 10 M
Question # 5 of 5 (Start time: 04:00:00 PM, 10 March 2022)
Given that the moment of inertia of the disc about a line passing through the center and perpendicular to the plane of disc is $\frac{1}{2}Mr^2$, then the mome
inertia of the disc about any diameter is
Select the correct option
Reload Mat
$\frac{1}{4}Mr^2$
$\frac{1}{2}Mr^2$


Search...


2:54
396 B/s
quiz.vu.edu.pk
Question # 1 of 5 (Start time: 02:54:43 PM, 10 March 2022)
Total Mi
The axis about which moment of area is taken is known as ——.
Select the correct option
Reload Math Equa
axis of area
axis of moment
axis of rotation
axis of reference


3:19
Quiz
https://quiz.vu.edu.pk
MC210203829: ZAIB UN NISA
Time Left 87 sec(s)
MTH622:Quiz no.4
Quiz Start Time: 03:19 PM, 10 March 2022
Question # 1 of 5 (Start time: 03:19:08 PM, 10 March 2022)
Total Marks: 1
Perpendicular axis theorem can be applied for which of the following bodies?
Select the correct option
Reload Math Equations
Disc of radius R and thickness t
A cube of side 'a'
Ring having radius R & negligible cross section
Cylinder of radius R and height h


3:20
Quiz
https://quiz.vu.edu.pk
MC210203829: ZAIB UN NISA
Time Left 87 sec(s)
MTH622:Quiz no.4
Quiz Start Time: 03:19 PM, 10 March 2022
Question # 2 of 5 (Start time: 03:20:47 PM, 10 March 2022)
Total Marks: 1
The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the difference of the moment
of inertia about two mutually perpendicular axis lying in the
Select the correct option
false
true

3:23

Quiz
https://quiz.vu.edu.pk

MC210203829: ZAIB UN NISA

Time Left 82 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 03:19 PM, 10 March 2022

Question # 4 of 5 (Start time: 03:23:05 PM, 10 March 2022)

Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- perpendicular
- coincident



Click to Save Answer & Move to Next Question

5:31 PM

quiz.vu.edu.pk/QuizQue

95

Question # 4 of 5 (Start time: 05:31:14 PM, 10 March 2022)

Point, where the total volume of the body is assumed to be concentrated is—.

Select the correct option

- center of area
- centroid of volume
- centroid of mass
- all of the mentioned

MTH622:Quiz no.4 Quiz Start Time: 04:50 PM, 10 March 2022

Question # 1 of 5 (Start time: 04:50:05 PM, 10 March 2022) Total Marks: 1

The principal moment of inertia matrix is a

Select the correct option

- identity matrix
- diagonal matrix

Click to Save Answer & Move to Next Question

4G 3G 3:23

Quiz
https://quiz.vu.edu.pk

MC210203829: ZAIB UN NISA

Time Left 88 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 03:19 PM, 10 March 2022

Question # 5 of 5 (Start time: 03:23:36 PM, 10 March 2022)

Total Marks: 1

Three mutually orthogonal eigenvectors for a 3×3 symmetric matrix exist if

Select the correct option

- the eigenvalues are only distinct
- the eigenvalues are imaginary
- the eigenvalues are either distinct or repeated ✓
- the eigenvalues are only repeated

Click to Save Answer & Move to Next Question